



GCSE – NEW

3445U20-1A



S17-3445U20-1A

APPLIED SCIENCE (Double Award)

UNIT 2: Space, Health and Life

**Pre-Release Article for use in the following examinations on
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Contributors to the greenhouse effect

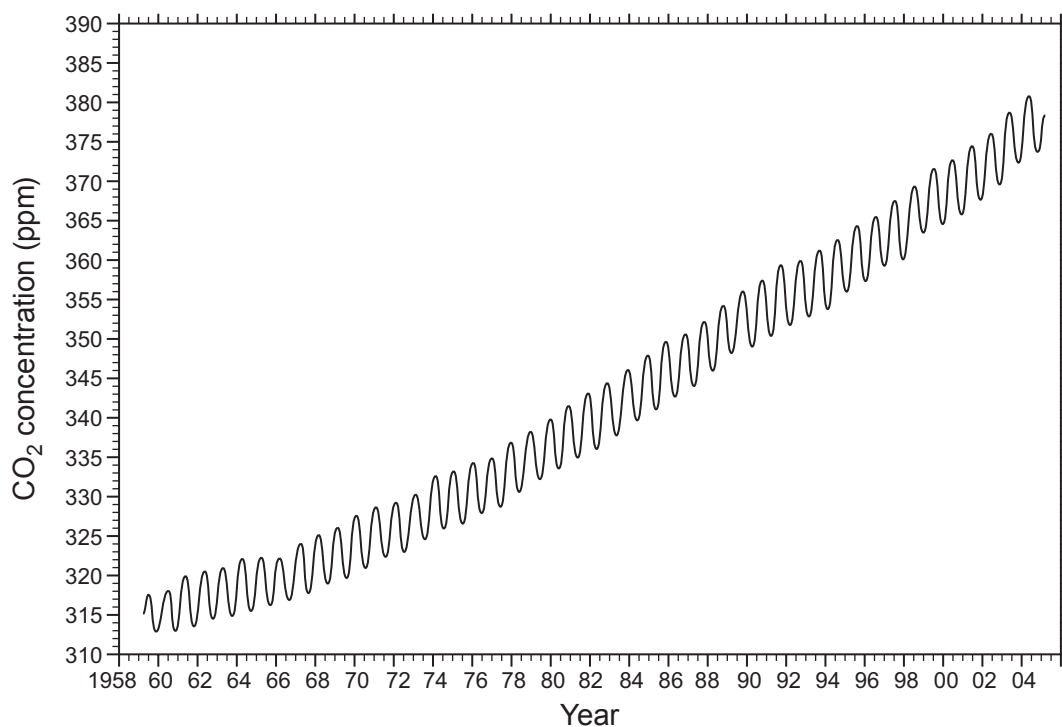
The main greenhouse gases emitted by human activities in the UK over time are shown in **Table 1**.

Table 1: UK greenhouse gas emission trends by gas, 1990-2014 in MegatonnesCO₂eq (MtCO₂eq)

	1990	1995	2000	2005	2010	2013	2014
net CO ₂ emissions	592.8	557.1	554.3	554.1	495.8	463.3	422.0
methane	137.0	130.4	114.4	92.1	66.2	55.8	53.5
nitrous oxide	49.4	40.0	29.6	25.6	22.5	21.4	21.9
hydrofluorocarbons (HFC)	14.4	19.1	9.8	13.1	16.6	16.0	16.3
perfluorocarbons	1.7	0.6	0.6	0.4	0.3	0.3	0.3
sulfur hexafluoride	1.3	1.3	1.8	1.1	0.7	0.5	0.5
Total greenhouse gases	796.6	748.5	710.6	686.3	602.1	557.3	514.4

Graph 1 shows seasonal CO₂ concentrations measured in Mauna Loa between 1958 and 2004. Each year there is a peak in November and a dip in May.

Graph 1: Monthly average carbon dioxide concentration at Mauna Loa Observatory, Hawaii.



On some websites it is claimed that the contribution of water vapour to the greenhouse effect is often ignored. **Table 2** was constructed from data published by the U.S. Department of Energy. It does not include water vapour. It refers to global atmospheric concentrations in the year 2000.

Table 2

greenhouse gas	concentration in the atmosphere (parts per billion (ppb))				
	pre-industrial baseline	natural additions	man-made additions	total	percent of total (%)
carbon dioxide	288 000	68 520	11 880	368 400	99.438
methane	848	577	320	1 745	0.471
nitrous oxide	285	12	15	312	0.084
CFC's	25	0	2	27	0.007
Total	289 158	69 109	12 217	370 484	100.000

Using corrections for the **greenhouse contribution** of each gas gives a more meaningful comparison of greenhouse gases, based on the conversion:

$$\text{greenhouse contribution} = (\text{concentration in the atmosphere}) \times (\text{multiplier})$$

Table 3: Contribution of some greenhouse gases to greenhouse effect

greenhouse gas	multiplier
carbon dioxide	1
methane	21
nitrous oxide	310

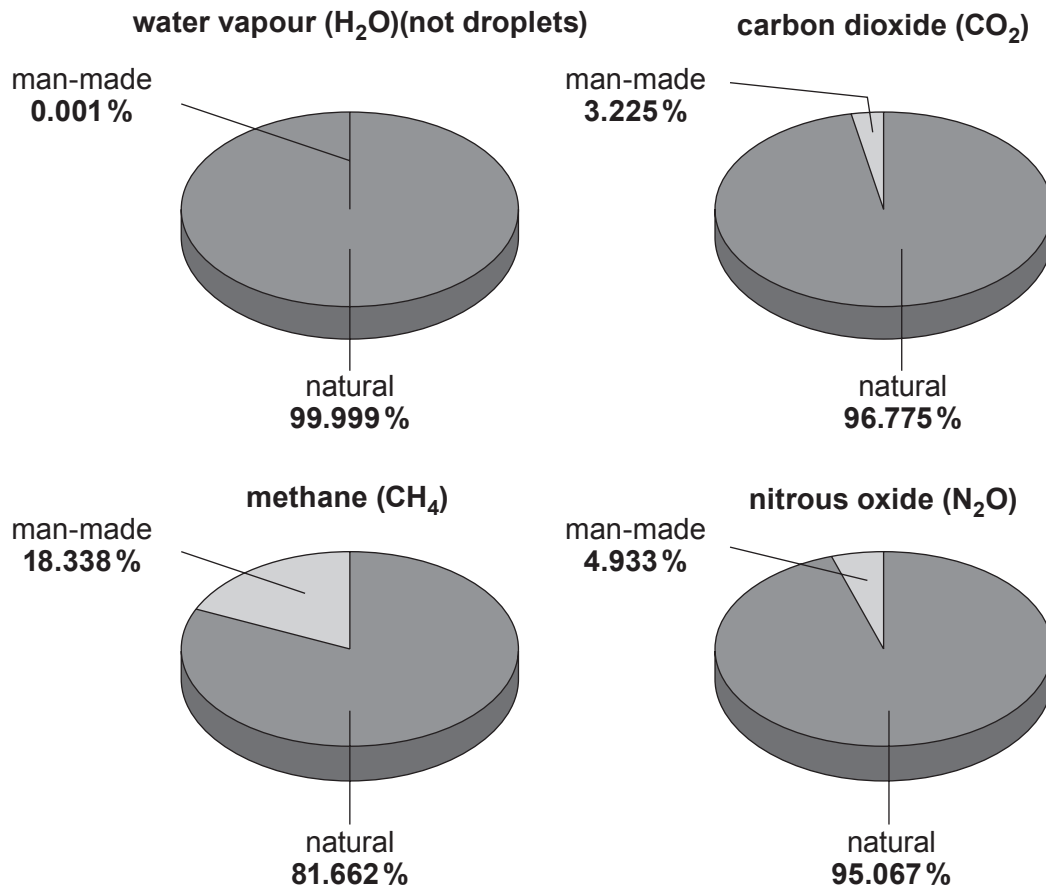
Table 4 shows what happens when the effect of water vapour is factored in.

Table 4

greenhouse gas	% of total	% of total including water vapour
water vapour		95.00
carbon dioxide	72.40	3.62
methane	7.10	0.36
nitrous oxide	19.00	0.95
CFC's (and other misc. gases)	1.40	0.07
total	100.00	100.00

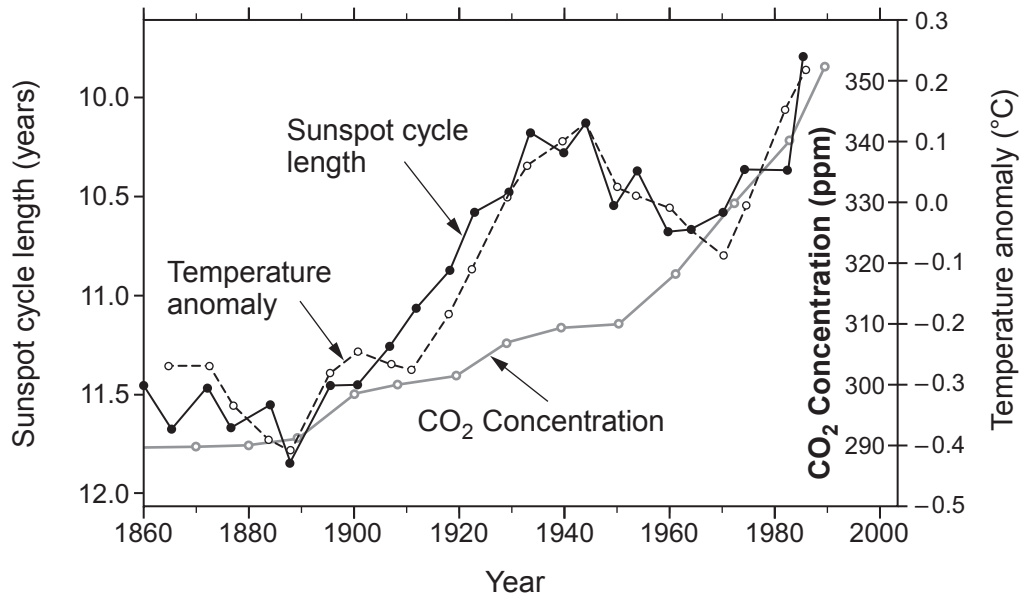
Other websites compare the human contribution to the greenhouse effect with natural contributions. The pie charts below show how much of each greenhouse gas is **natural** and how much is **man-made**.

Chart 1

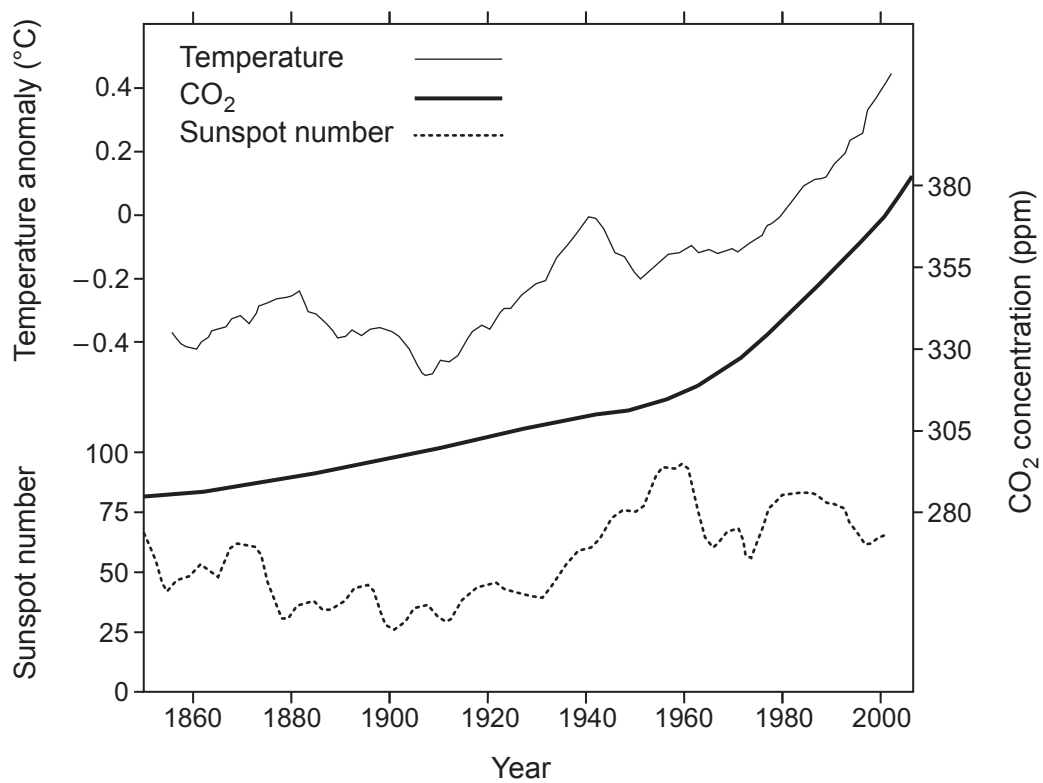


Some websites claim that global temperature depends on the activity of the Sun. The graphs below are taken from two different websites. The temperature anomaly refers to variations in air temperature.

Graph 2



Graph 3





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3445U20-1A



S18-3445U20-1A

APPLIED SCIENCE (Double Award)

UNIT 2: Space, Health and Life

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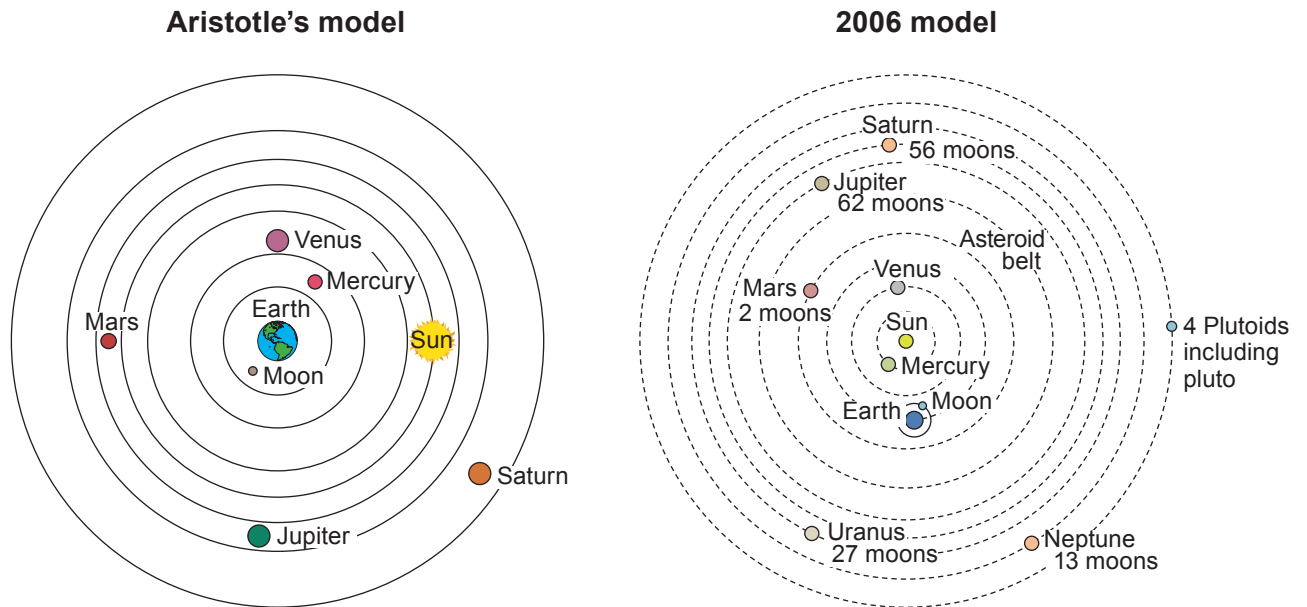
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History of our Solar System

There are different models of the Solar System. An ancient model was suggested by Aristotle. Modern models of the Solar System have changed from Aristotle's model.

Diagram 1: Models of the Solar System



Diagrams not drawn to scale

Properties of some objects in the Solar System are described in Table 1.

Table 1: Properties of objects in our Solar System

	Mercury	Venus	Earth	Moon	Mars	Jupiter	Saturn	Pluto
Mass (10^{24} kg)	0.330	4.87	5.97	0.073	0.642	1898	568	0.015
Diameter (km)	4 879	12 104	12 756	3 475	6 792	142 984	120 536	2 370
Density (kg/m^3)	5 427	5 243	5 514	3 340	3 933	1 326	687	2 095
Gravity (m/s^2)	3.7	8.9	9.8	1.6	3.7	23.1	9.0	0.7
Length of day (hours)	4 222.6	2 802.0	24.0	708.7	24.0	9.9	10.7	153.3
Distance from Sun (AU)	0.4	0.7	1	1	1.5	5.2	9.5	39.5
Orbital period (days)	88.0	224.7	365.2	27.3	687.0	4 331	10 747	90 560
Orbital velocity (km/s)	47.4	35.0	29.8	1.0	24.1	13.1	9.7	4.7
Mean temperature ($^{\circ}\text{C}$)	167	464	15	-20	-65	-110	-140	-225
Ring system?	No	No	No	No	No	Yes	Yes	No
Atmosphere	None	CO_2	$\text{N}_2 + \text{O}_2$	None	CO_2	$\text{H}_2 + \text{He}$	$\text{H}_2 + \text{He}$	N_2

The Sun's activity

The solar magnetic field changes on an eleven year cycle. Every solar cycle, the number of sunspots, flares, and solar storms increases to a peak, which is known as the solar maximum. Then, after a few years of high activity, the Sun will enter a few years of low activity, known as the solar minimum. This pattern is called the sunspot cycle. Sunspot cycles have been tracked since 1755.

Graph 1 shows the number of sunspots during different sunspot cycles 18 to 23 since 1940.

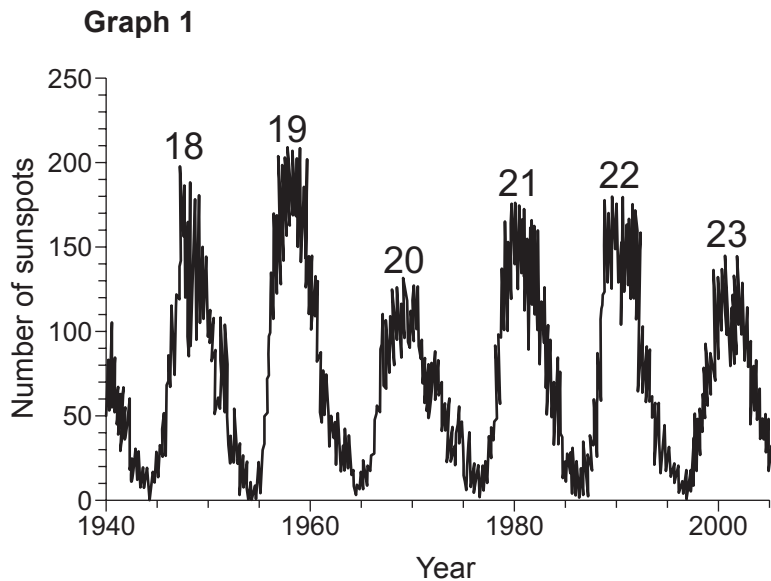
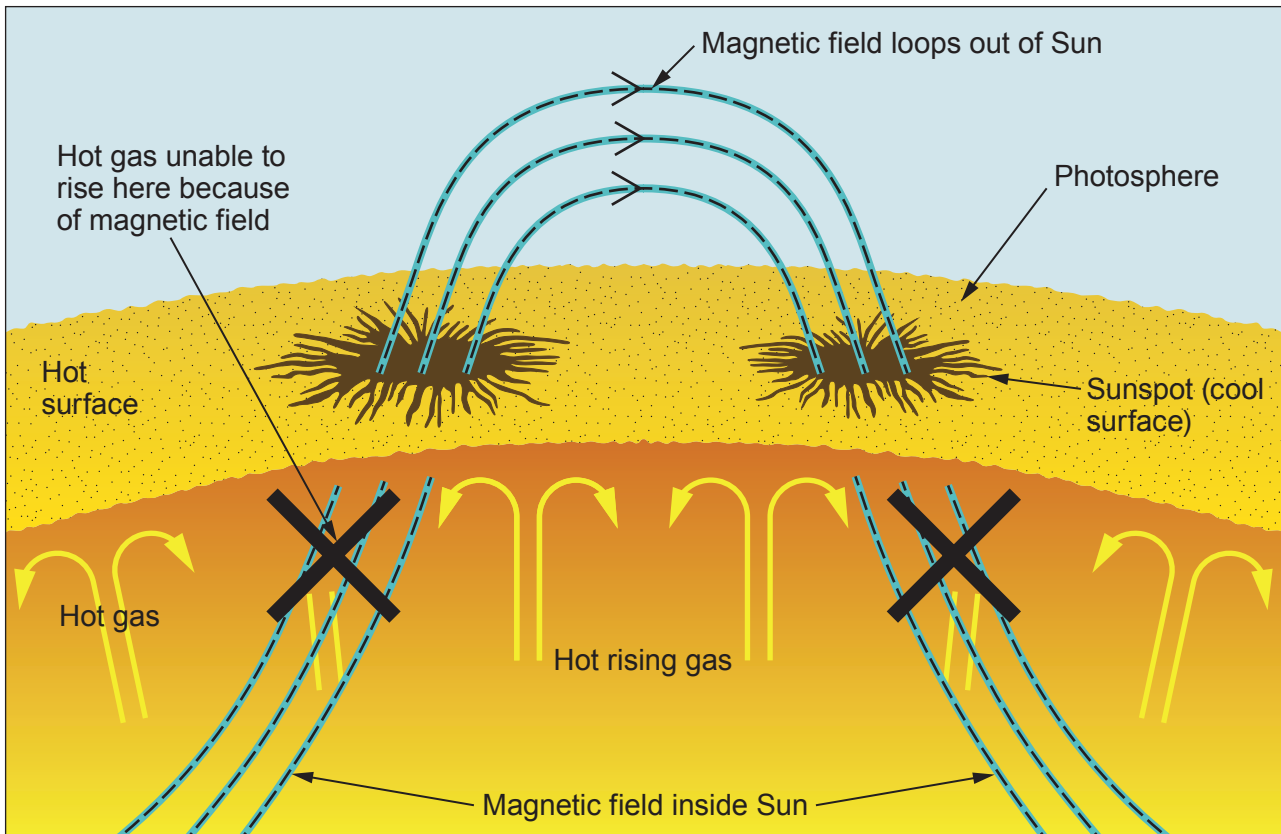


Diagram 2 shows the magnetic field near sunspots.

Diagram 2

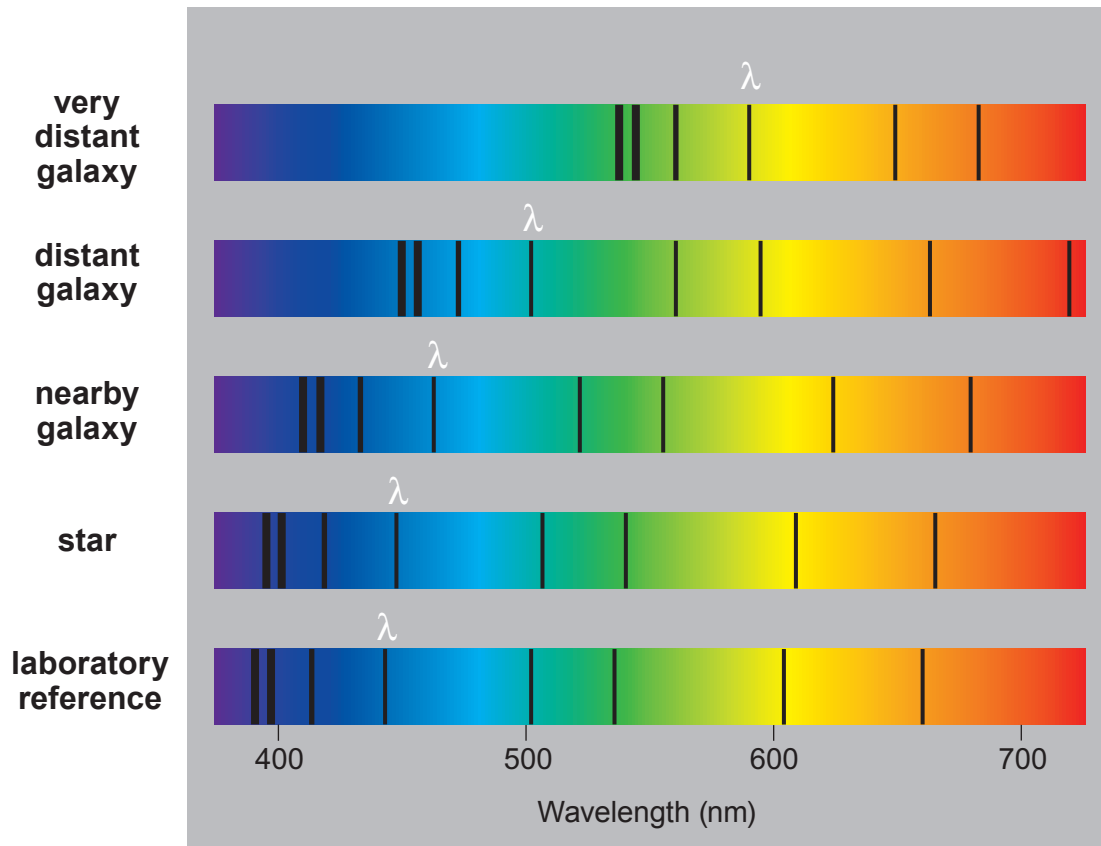


A solar flare is an intense burst of radiation coming from the release of magnetic energy associated with sunspots.

Spectra

The light emitted from stars and galaxies is crossed by dark lines in a regular pattern. Some of these spectra are shown in **Diagram 3**.

Diagram 3



The speed of em waves (300 000 km/s) is related to frequency and wavelength by the equation:

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$



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S19-3445U20-1A

APPLIED SCIENCE (Double Award)

UNIT 2: Space, Health and Life

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Factors affecting Human Health

Smoking, drinking alcohol and diet are lifestyle choices that affect health.

1. Smoking

Figure 1: What happens to the body in the time after smoking the last cigarette

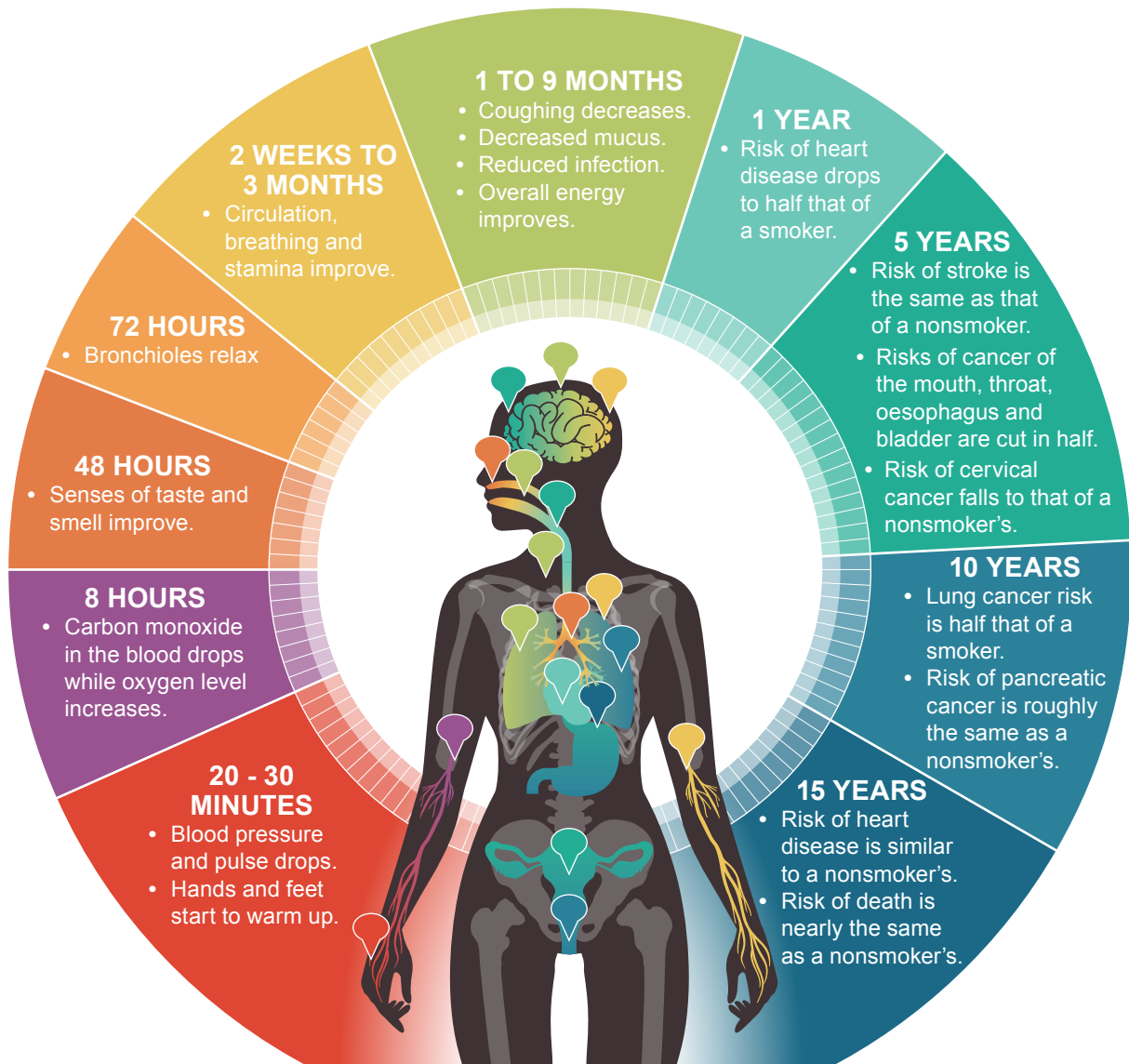


Figure 2: Percentage of the Welsh population who were current smokers and the percentage of smokers who have quit

Year	Percentage	
	Current smokers	Smokers who have quit
1974	45.6	26.7
1976	41.8	31.0
1982	35.3	38.8
1984	34.0	40.2
1986	32.7	42.7
1990	30.0	45.3
1992	28.4	47.7
1994	26.8	48.8
1996	28.0	47.2
2000	27.0	46.5
2002	25.9	48.1
2006	22.0	51.9
2008	21.6	54.7
2010	20.3	54.9
2012	20.4	50.2
2014	18.8	54.6
2016	17.8	56.7

2. Alcohol

The Chief Medical Officer's guideline for both men and women is not to drink more than 14 units a week on a regular basis.

Figure 3: Maximum recommended weekly intake of alcoholic drinks



The NHS defines binge drinking as “drinking lots of alcohol in a short space of time or drinking to get drunk”. This amounts to drinking more than 6 units of alcohol in one session. The body can only process one unit of alcohol per hour.

3. Diet

A healthy diet should consider the guideline daily amounts (GDA) shown in the table below and balance this against personal energy requirements (PER).

Figure 4: Guideline daily amounts (GDA)

Guideline Daily Amount (GDA)			
Typical values	Women	Men	Children (5-10 years)
Calories	2 000 kcal	2 500 kcal	1800 kcal
Protein	45 g	55 g	24 g
Carbohydrate	230 g	300 g	220 g
Sugars	90 g	120 g	85 g
Fat	70 g	95 g	70 g
of which are saturates	20 g	30 g	20 g
Fibre	24 g	24 g	15 g
Salt	6 g	6 g	4 g

Figure 5: Example of a food label on a variety pack of cereals

Nutrition Facts	Wheat Squares Sweetened	Corn Flakes Not Sweetened	Mixed Grain Flakes Sweetened
1 serving per container Serving size 1 box	(35g)	(19g)	(27g)
Amount per serving Calories	130 kcal	70 kcal	100 kcal
Total Fat	0g	0g	0g
of which are saturates	0mg	0mg	0mg
Salt	0mg	200mg	120mg
Total Carbohydrate	29g	17g	24g
Dietary Fibre	3g	1g	1g
Total Sugars	8g	8g	13g
Protein	4g	1g	1g
Vitamin D	0.002mg	0.002mg	0mg
Calcium	0mg	0mg	0mg
Iron	2mg	1mg	4mg
Potassium	125mg	25mg	30mg

People have different personal energy requirements (PER) as shown by the equation below.

$$\begin{array}{ccccc} \text{Personal Energy} & = & \text{Basic Energy} & + & \text{Extra Energy} \\ \text{Requirement} & & \text{Requirement} & & \text{Requirement} \\ \text{(PER)} & & \text{(BER)} & & \text{(EER)} \end{array}$$

The BER is given by:

$$\text{BER} = 32 \times \text{body mass}$$

and the EER is given by:

$$\text{EER} = 8.5 \times \text{number of hours training} \times \text{body mass}$$

Poor diet can lead to obesity or malnutrition. This affects the Body Mass Index (BMI). BMI is calculated using the equation:

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

The risk of developing health problems is related to BMI values as shown in **Figure 6**.

Figure 6

Classification	BMI Category (kg/m^2)	Risk of developing health problems
Underweight	< 18.5	Increased
Normal weight	18.5 – 24.9	Lowest
Overweight	25.0 – 29.9	Increased
Obese class I	30.0 – 34.9	High
Obese class II	35.0 – 39.9	Very high
Obese class III	≥ 40.0	Extremely high

Figures 7 and 8 below give BMI values for people of different masses or heights.

Figure 7

Mass (kg)	Height (m)	Height ² (m ²)	$\frac{1}{\text{height}^2}$ (/m ²)	BMI (kg/m ²)
40	1.7	2.9	0.35	14
50	1.7	2.9	0.35	17
60	1.7	2.9	0.35	21
70	1.7	2.9	0.35	24
80	1.7	2.9	0.35	28
90	1.7	2.9	0.35	31
100	1.7	2.9	0.35	35

Figure 8

Mass (kg)	Height (m)	Height ² (m ²)	$\frac{1}{\text{height}^2}$ (/m ²)	BMI (kg/m ²)
65	1.2	1.4	0.69	45
65	1.3	1.7	0.59	38
65	1.4	2.0	0.51	33
65	1.5	2.3	0.44	29
65	1.6	2.6	0.39	25
65	1.7	2.9	0.35	22
65	1.8	3.2	0.31	20
65	1.9	3.6	0.28	18

Obesity in people with relatively large BMI values can cause the development of diabetes. However maturity onset diabetes of the young (MODY) is a rare form of diabetes which is different from both Type 1 and Type 2 diabetes, and runs in families. MODY is caused by a mutation in a single gene making it a dominant allele.



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APPLIED SCIENCE (Double Award)

UNIT 2: Space, Health and Life

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The Red Admiral butterfly

Red Admiral is the common name for one species of butterfly that visits the UK.



It has the scientific name *Vanessa atalanta*. Its classification is shown in **Table 1**.

Table 1: Scientific classification of the Red Admiral

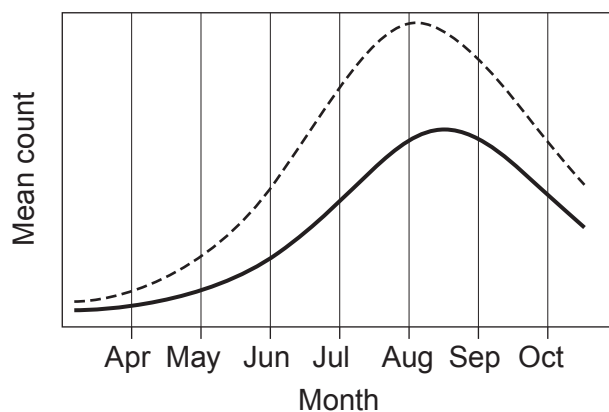
Scientific classification	
Kingdom:	Animalia
Phylum:	Arthropoda
Class:	Insecta
Order:	Lepidoptera
Family:	Nymphalidae
Genus:	<i>Vanessa</i>
Species:	<i>atalanta</i>
Scientific name:	<i>Vanessa atalanta</i>

Adult Red Admirals may be seen in the UK throughout the year. There is a seasonal increase in number in May and June as more travel from southern Europe. The Red Admirals breed and give rise to the next generation of adults which peak between mid-August and early October. They breed once a year. In autumn the offspring travel southwards.

The variation in the adult population at Oxwich Bay is shown in **Figure 1** below.

The solid line gives the mean count from 1976 to 2019, and the dashed line gives the mean count for 2019.

Figure 1



Food webs

Table 2 shows the feeding relationships in a typical woodland habitat.

Table 2

Food source	Eaten by
berries	greenfly, butterfly, titmouse, grasshopper
plantain	grasshopper, mouse, rabbit
greenfly	ladybird
butterfly	frog
grasshopper	frog
titmouse	snake, buzzard, fox
mouse	buzzard, fox
rabbit	buzzard, fox
ladybird	dragonfly, titmouse
dragonfly	frog
frog	snake
snake	buzzard

Food chains and a food web can be drawn using this information. The food web shows how each living thing is linked to the others.

The living things are also grouped into **trophic levels**.

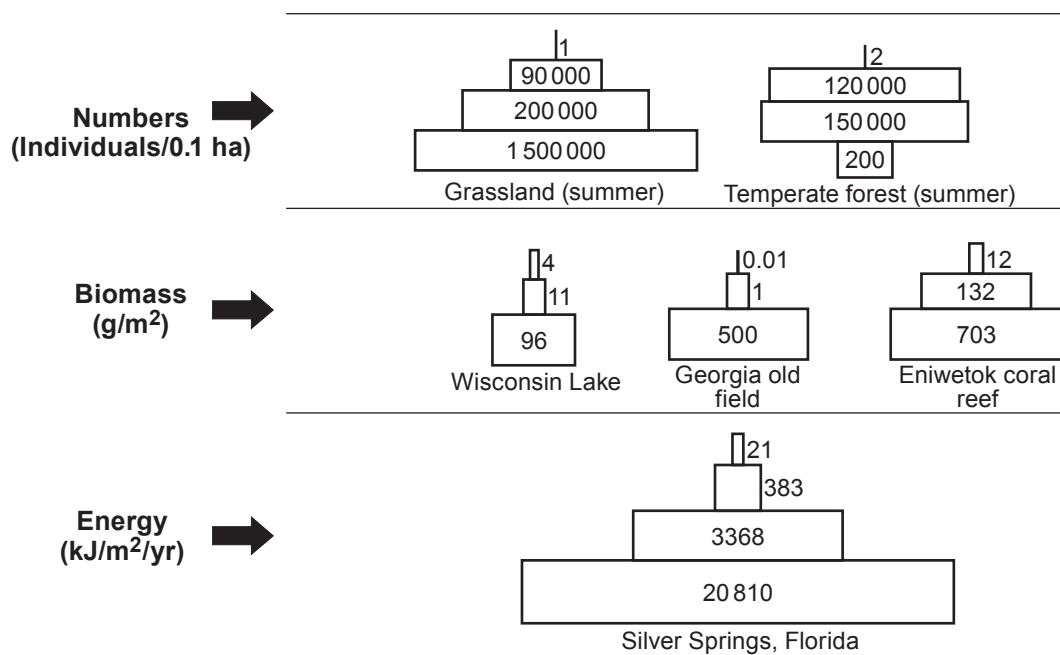
In the simplest scheme, the first trophic level is usually plants, the second trophic level is usually herbivores, and subsequent levels are carnivores. In a food web some organisms can appear in more than one trophic level.

Ecological pyramids

These are pyramid-shaped diagrams representing the numbers of organisms, energy levels, and biomass of an ecosystem. Values are usually high for the lowest trophic level and low for the highest trophic level.

Figure 2 shows different types of ecological pyramids from various habitats throughout the world.

Figure 2: Ecological pyramids



The efficiency of transferring energy between trophic levels can be calculated using the equation

$$\% \text{ efficiency} = \frac{\text{useful energy at a trophic level}}{\text{available energy at previous trophic level}} \times 100$$



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APPLIED SCIENCE (Double Award)

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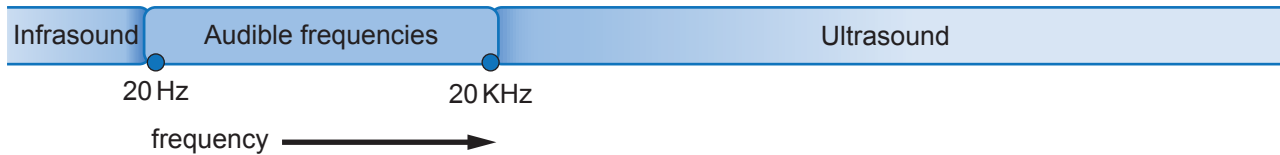
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Diagnosis and treatment.

Different types of waves are used in the diagnosis and treatment of illness. One of these types is ultrasound. The position of ultrasound on the sound spectrum is shown in **Figure 1**.

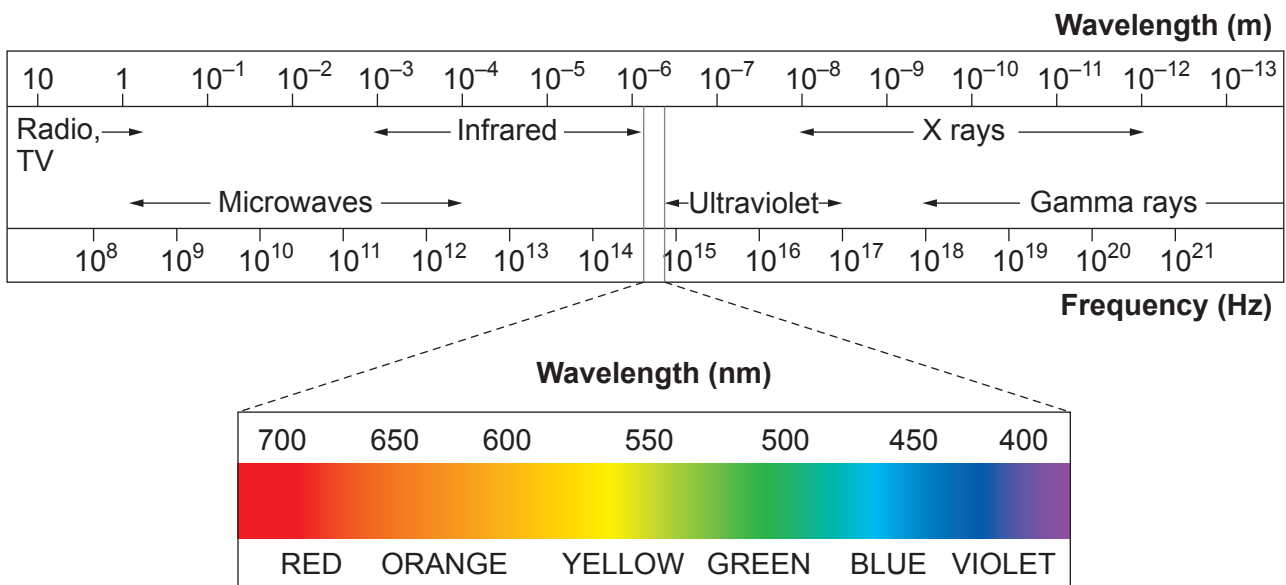
Figure 1



Medical ultrasound scans use frequencies in the range 2 MHz to 20 MHz.
(1 MHz = 1 000 000 Hz)

The electromagnetic (em) spectrum is a family of waves shown in **Figure 2**. Some em waves have medical uses.

Figure 2



A hospital radiology department displays information about the dose a patient receives from different types of X-rays. This is shown in **Figure 3**.

Figure 3

Type of X-Ray	Received dose in units	Equivalent days of background radiation
knee	1	1.5
chest	2	3
skull	10	15
spine	100	150
hip	30	45
pelvis	100	150
abdomen	150	225

Radioisotopes

Radioisotopes are used in many medical applications, including diagnosis and therapy.

Radioactive tracer – a radioisotope inserted into the body and used to track a substance by monitoring its radioactive emissions.

Radiation therapy – use of radiation to damage the DNA of cancer cells, which kills them or keeps them from dividing. There are two types.

- **External beam radiation therapy** – radiation delivered by a machine outside the body to kill cancer cells.
- **Internal radiation therapy (brachytherapy)** – radiation from a radioactive substance introduced into the body to kill cancer cells.

The table in **Figure 4** gives information about some radioisotopes.

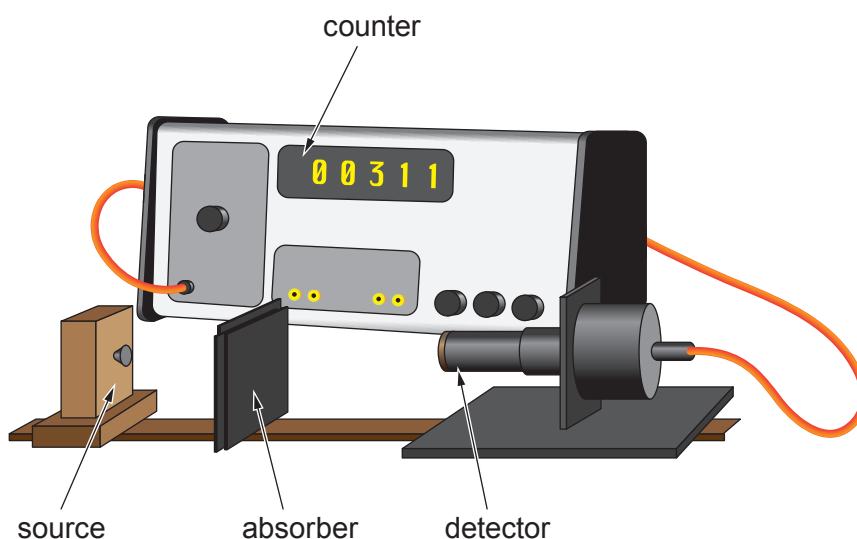
Figure 4

Radioisotope	Half-life	Decay mode
cobalt-55	17.3 hours	beta
cobalt-58	70 days	beta
cobalt-60	5.3 years	beta and gamma
francium-220	27.4 seconds	alpha
gold-198	2.7 days	beta
iodine-123	13 hours	gamma
iodine-125	59.4 days	gamma
iodine-128	25 minutes	beta
iodine-131	8.4 days	beta and gamma
lutetium-177	6.7 days	beta and gamma
radium-226	1 600 years	alpha
radon-222	3.8 days	alpha
scandium-47	3.3 days	beta
technetium-99m	6 hours	gamma
thallium-201	73 hours	gamma
yttrium-86	14.7 hours	beta
yttrium-88	107 days	beta

Identifying radioisotopes

The apparatus in **Figure 5** is used to identify the type(s) of radiation emitted by a radioisotope.

Figure 5



This apparatus was used to analyse a number of different radioisotopes, **R1 – R6**. The results obtained from different radioisotopes are shown in the table in **Figure 6**. The data has been corrected for background.

Figure 6

Radioisotope	Corrected count rate (units)			
	no absorber	paper	aluminium	lead
R1	250	250	250	55
R2	200	120	120	40
R3	125	0	0	0
R4	175	175	80	20
R5	75	75	0	0
R6	100	75	0	0

Modelling half-life

Students model the random nature of radioactive decay using 6-sided dice.
Each group of students is given 50 dice.
There are 12 groups in total.

Model 1

1. They throw the dice and remove any that land with the number 6 facing upwards.
2. They throw the remaining dice again.

Figure 7 shows the results for Group A and the whole class.

Figure 7

Throw	Number of dice remaining with Model 1	
	For Group A	For the class
0	50	600
1	41	505
2	34	420
3	27	350
4	23	290
5	19	240
6	17	200
7	13	170
8	9	140

Model 2

1. They throw the dice and remove any that land with the numbers 5 and 6 facing upwards.
2. They throw the remaining dice again.

Model 3

1. They throw the dice and remove any that land with the numbers 4, 5 and 6 facing upwards.
2. They throw the remaining dice again.

Figure 8 shows the results for the whole class for **Models 2** and **3**.

Figure 8

Throw	Number of dice remaining in the class	
	Model 2	Model 3
0	600	600
1	400	310
2	265	145
3	175	80
4	115	40
5	75	15
6	45	10
7	25	5
8	20	3

Each model represents a different radioisotope with a different half-life which can be calculated from the data.